Pythia-14M: matched warmup-only learning rates
Identical initialization and data order; seed 1234; final partial batches omitted

Raw gradient

Adam stored momentum m

Adam adaptive direction

Adjacent cosine (25-update median)

Cumulative B/A

Current norm (log scale)

Optimizer update

No decay, LR 0.001

No decay, LR 0.0001

No decay, LR 0.01